Entropic regularization

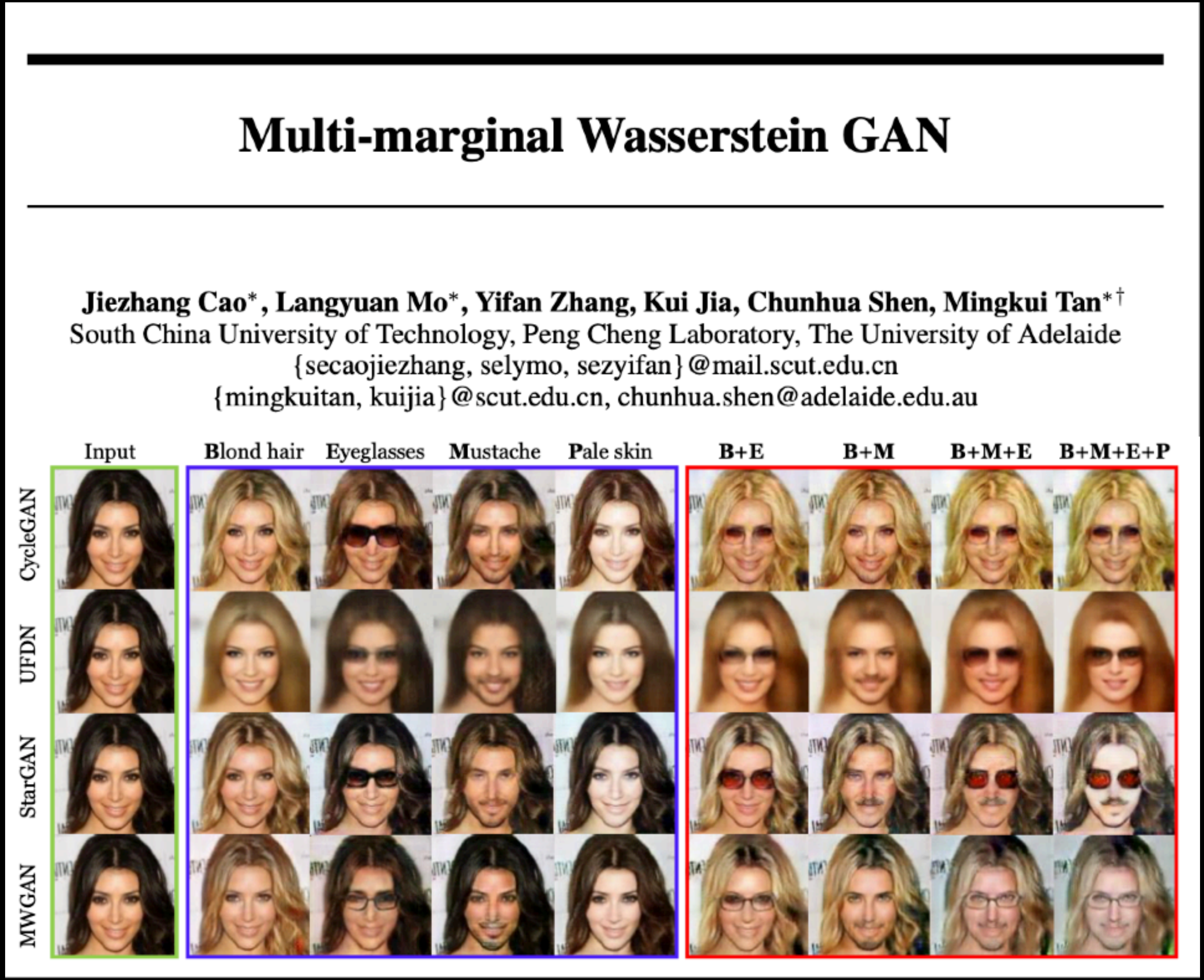
Multi-Marginal OT: Computation

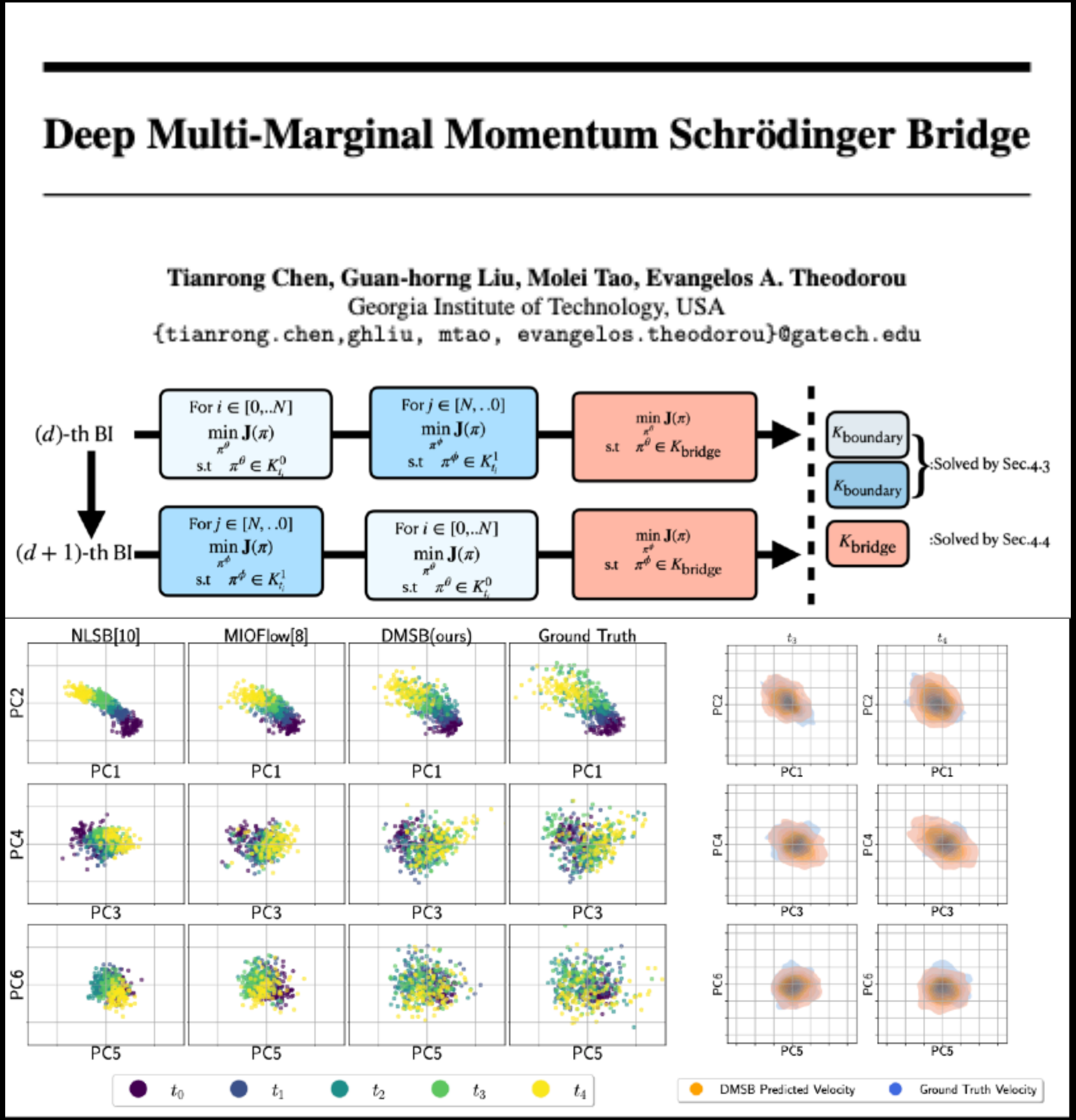
- Computational challenge: combinatorial in S (number of marginals)
- **Entropic regularization** naturally extends to this setting: $\min_{\pi \in \Pi(\mathbf{a}_1,\ldots,\mathbf{a}_S)} \langle \pi, \mathbf{C} \rangle - H(\pi)$
- Yields unique solution of the form: $\pi_i = \mathbf{K}_i \prod_{s=1}^{S} (\mathbf{u}_s)_{i_s}$, where i is a multi-index vector $i = (i_1, \ldots, i_S)$
- Generalized Sinkhorn updates on S scaling vectors $\mathbf{u}_1, \ldots, \mathbf{u}_S$:

$$(\mathbf{u}_s)_{i_s} \leftarrow \frac{(\mathbf{a}_s)_{i_s}}{\sum_{l \neq s} \mathbf{K}_{i_1,\ldots,i_S} \prod_{r \neq s} (\mathbf{u}_l)_{i_r}}, \qquad s = 1,\ldots,S\,.$$

- Or in tensor form: $\mathbf{u}_s \leftarrow \mathbf{a}_s \oslash \left(\mathbf{K}\, \bar{\times}_s \left(\otimes_{r \neq s} u_r\right)\right), \qquad s = 1,\ldots,S\,.$

Multi-Marginal OT: Applications





Also: quantum chemistry (DFT), fluid mechanics